



USB-C diode-OR -> SY8089 buck (2.7-5.5Vin,2A) -> 3.3V. EN tied +5V. Vout=0.6*(1+R/R)=3.32V.

CC1
5.1k
GND

CC2
5.1k
GND

VBUS

D1
+5V
SS54
VBUS

C1
+5V
100uF
GND

C2
+5V
100nF
GND

U2
IN U2 LX
EN
GND
SW

C3
+5V
22uF
GND

SHIELD
A1
GND

VBUS
A4
VBUS
A5
CC1
B5
CC2
A7
USB_DM_RAW
B7
USB_DM_RAW
A6
USB_DP_RAW
B6
USB_DP_RAW
A8
SBU1
B8
SBU2

VBUS
5
USB_DP_RAW
4
USB_DM_RAW
3
USB_DP_RAW
2
GND

D2
+5V
SS54
EXT5V

Title: Power - USB-C / diode-OR / SY8089 buck 3.3V		
Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.3		Id: 2/7

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[illegible]

Title: ESP32-S3-WROOM-1-N8		
Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.3		Id: 3/7



PhotoMOS isolated dry contact x8. GPIO->330R->AQY212GS. HIGH=closed. <=60V/<=0.5A.

OUT1

OUT1_LED

OUT1_A

OUT1_B

OUT2

OUT2_LED

OUT2_A

OUT2_B

OUT3

OUT3_LED

OUT3_A

OUT3_B

OUT4

OUT4_LED

OUT4_A

OUT4_B

OUT5

OUT5_LED

OUT5_A

OUT5_B

OUT6

OUT6_LED

OUT6_A

OUT6_B

J13

J14

J15

J16

J17

J18

OUT1_A

OUT1_B

OUT2_A

OUT2_B

OUT3_A

OUT3_B

OUT4_A

OUT4_B

OUT5_A

OUT5_B

OUT6_A

OUT6_B

Sheet: /PhotoMOS/
File: photomos.kicad_sch
Title: 8x PhotoMOS dry-contact outputs
Size: A3 Date: Rev: 0.1
KiCad E.D.A. 10.0.3 Id: 5/7



Sheet: /CapInject/ File: capinject.kicad_sch		
Title: 4x capacitive injection outputs		
Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.3		Id: 7/7